In [24]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, roc_auc_score

# 1. Load Dataset
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target

# 2. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

# 3. Decision Tree Classifier
dt_model = DecisionTreeClassifier(max_depth=4, random_state=42)
dt_model.fit(X_train, y_train)
dt_preds = dt_model.predict(X_test)

# 4. Random Forest Classifier
rf_model = RandomForestClassifier(n_estimators=100, max_depth=4, random_state=42)
rf_model.fit(X_train, y_train)
rf_preds = rf_model.predict(X_test)

# 5. Extract Feature Importance
importances = rf_model.feature_importances_
indices = np.argsort(importances)[::-1]
print('Top 5 Features in Random Forest:')
for f in range(5):
    print(f'{f+1}. {X.columns[indices[f]]}: {importances[indices[f]]:.4f}')
```

```
Top 5 Features in Random Forest:
1. worst area: 0.1617
2. worst concave points: 0.1481
3. mean concave points: 0.1111
4. worst radius: 0.0777
5. mean concavity: 0.0709
```

In [23]:
```
!jupyter nbconvert --to html /content/smld_8_new.ipynb
```

```
[NbConvertApp] Converting notebook /content/smld_8_new.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 1 image(s).
[NbConvertApp] Writing 358509 bytes to /content/smld_8_new.html
```